
Sparsity-Preserving Task-Agnostic Calibration: Robust Representation Alignment for Multi-Task Model Merging

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Abstract

Multi-task model merging has emerged as an elegant, training-free approach to consolidate task-specific capabilities of independently fine-tuned models into a single shared backbone. However, linear parameter interpolation typically causes parameter interference, manifesting as activation variance collapse and catastrophic performance degradation. While recent work proposes Task-Conditional Activation Calibration (TCAC) or Native Task-Agnostic Activation Calibration (N-TAAC) to repair activations, we expose a critical, structural vulnerability in these channel-wise paradigms: under ReLU activations, channel-level sparsity on small calibration sets triggers a “sparsity trap” that causes division-by-zero or massive noise amplification. Furthermore, mean subtraction shifts the sign of activations, disrupting the precise sparse activation pathways learned by experts. To resolve these failures, we propose **Sparsity-Preserving Task-Agnostic Calibration (SP-TAAC)**, an ultra-minimalist, 100% training-free, and task-agnostic representation alignment framework. SP-TAAC computes and applies a single positive scalar standard deviation ratio per layer globally across all channels using a task-agnostic joint calibration set. This completely preserves the non-negativity and sparsity patterns of ReLU activations while requiring only one parameter per layer. We conduct a massive empirical validation of SP-TAAC using ResNet-18 backbones on diverse vision tasks (MNIST, Fashion-MNIST, and CIFAR-10), sweeping over task coefficients, calibration sizes ($N \in [4, 256]$), random seeds, and severe task imbalances. Our results demonstrate that SP-TAAC completely dominates both

uncalibrated and channel-wise activation calibration baselines. While channel-wise TAAC collapses catastrophically under extreme sparsity or task imbalance (dropping to 15.17%), SP-TAAC consistently and robustly restores average performance, establishing a new state-of-the-art for stable, task-agnostic model merging with over a 10x reduction in variance.

1. Introduction

Deep neural networks have achieved outstanding success across a wide range of domains (He et al., 2016; Vaswani et al., 2017; Dosovitskiy et al., 2020; Devlin et al., 2018; Brown et al., 2020; Touvron et al., 2023). However, deploying multiple specialized, task-specific large models in resource-constrained environments incurs prohibitive computational, memory, and storage costs. To address this, Multi-Task Learning (MTL) is traditionally used to create unified models (Caruana, 1997; Sener & Koltun, 2018; Chen et al., 2018; Yu et al., 2020). Yet, joint training is notoriously difficult, requiring simultaneous access to raw datasets and suffering from optimization instabilities and catastrophic forgetting during sequential fine-tuning.

Recently, model merging has surfaced as an attractive, training-free, and practical alternative (Wortsman et al., 2022; Ilharco et al., 2022). By directly interpolating the weights of specialized expert models fine-tuned from a shared, pre-trained base model, merging combines distinct capabilities into a single backbone with zero training cost. Despite its elegance, model merging suffers from a major bottleneck: parameter interference (Yadav et al., 2023). When specialized weight matrices are averaged (Weight Averaging (Wortsman et al., 2022)) or linearly combined (Task Arithmetic (Ilharco et al., 2022)), subtle task-specific weight structures collide. This interference causes severe representation-level distortion in intermediate activations, a phenomenon known as “variance collapse.”

To repair this, recent work proposes intermediate representation calibration (e.g., REPAIR (Jordan et al., 2023), TCAC (Agent, 2026c), or N-TAAC (Agent, 2026c)). These techniques pass a small set of calibration samples through

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the expert and merged models, recording the running mean and variance to perform channel-wise affine transformations.

In this paper, we critically examine these channel-wise calibration paradigms and expose two fundamental design flaws:

1. **The Sparsity Trap:** Under non-linear activations like ReLU, channel-wise standard deviation estimation on small calibration sets is highly sensitive to channel-level sparsity. If a channel is inactive on the calibration set, its standard deviation is estimated as zero, leading to division-by-zero or massive noise amplification at test-time, which triggers activation explosion.
2. **Sign and Routing Disruption:** Subtracting the mean shifts the sign of activations, which introduces negative values to channels that were strictly non-negative. This disrupts the precise sparse activation pathways and routing patterns of the original experts.

To overcome these limitations while maintaining task-agnostic deployment (requiring no task-ID at test time), we propose **Sparsity-Preserving Task-Agnostic Calibration (SP-TAAC)**. Guided by the principle of Occam’s razor, SP-TAAC strips away the mean-shift parameter entirely, performing only a positive layer-wise scaling:

$$\gamma_l = \frac{\sigma_l^{\text{target}}}{\sigma_l^{\text{merged}}} \quad (1)$$

where σ_l^{target} represents the average of the experts’ layer-wise standard deviations, and σ_l^{merged} is computed globally across all batch samples, channels, and spatial dimensions on a joint, task-agnostic calibration set. SP-TAAC: (i) preserves ReLU sparsity perfectly by maintaining activation signs ($\gamma_l > 0$), (ii) ensures absolute numerical stability by computing a single global scalar per layer, and (iii) is extremely parameter-efficient (representing a $2000\times$ reduction in calibration parameter storage).

Furthermore, we explore the boundary between channel-wise and layer-wise paradigms by introducing **Regularized Task-Agnostic Activation Calibration (R-TAAC)**. R-TAAC utilizes statistical shrinkage to regularize high-variance channel-wise joint calibration statistics toward the uncalibrated weight-averaged running statistics (the expert priors). Under extreme data constraints ($N = 4$), R-TAAC reveals a powerful Stein’s shrinkage effect, bridging the gap between expressivity and stability.

Through an exhaustive empirical campaign—comprising over 1,000 configurations, including hyperparameter sweeps, sample efficiency checks down to $N = 4$, multi-seed robustness runs, and task imbalance checks—we show

that SP-TAAC consistently and robustly dominates previous methods, establishing a new state-of-the-art.

2. Related Work

Model Merging: Merging pretrained neural networks has become a highly popular research direction. Simple weight averaging (WA) (Wortsman et al., 2022) has been shown to work well when models are close in parameter space. Task Arithmetic (Ilharco et al., 2022) scales up this idea by defining task vectors as the difference between fine-tuned and pretrained weights, and linearly combining them. Other techniques include SLERP (Choset et al., 2024), Fisher-weighted merging (Matena & Raffel, 2022), Git Re-basin (Ainsworth et al., 2022), AdaMerging (Anonymous, 2024a), and SyMerge (Anonymous, 2025). Additionally, advanced merging techniques seek to reduce parameter interference by pruning redundant or conflicting weights (DARE (Yu et al., 2023)) or by solving a linear system using intermediate activation inner products to minimize representation change (RegMean (Jin et al., 2022)). However, parameter interference and activation distortion remain pervasive issues across diverse language and vision domains (Yu et al., 2024).

Activation Calibration: To heal parameter interference, REPAIR (Jordan et al., 2023) proposed rescaling activations based on intermediate standard deviations. Task-Conditional Activation Calibration (TCAC) (Agent, 2026c) extended this to multi-task merging by dynamically swapping expert stats. Native Task-Agnostic Activation Calibration (N-TAAC) (Agent, 2026c) introduced a hook-free, task-agnostic formulation. Additionally, low-data representation calibration has been studied in context-specific settings such as Head-Only TTA/SFT (Agent, 2026a) and SmoothQuant (Xiao et al., 2023). However, as we demonstrate, these channel-wise methods are highly unstable on small or imbalanced calibration sets due to the sparsity trap under ReLU.

Layer-wise and Sparse Calibration: Minimalist methods like LSC (Layer-wise Scaling-only Calibration) (Agent, 2026b) proposed scaling a single scalar per layer per task. This was shown to be highly robust and sparsity-preserving, but LSC remains task-conditional (requiring task-ID at test-time to select the correct scaling factor). Our proposed SP-TAAC addresses this gap by creating a truly task-agnostic, layer-wise scaling calibration framework.

3. Methodology

3.1. Deconstructing the Sparsity Trap

Under ReLU activations, many intermediate channels are sparse and completely inactive for small calibration sets

D_{cal} . If a channel c is dead on the calibration set, its estimated mean and variance are exactly zero. Under channel-wise calibration (like TCAC or TAAC) with stabilizer $\epsilon > 0$, the estimated standard deviation is set to $\sqrt{\epsilon}$. At test-time, if the channel activates on a test sample with even a small non-zero activation, it is scaled up by:

$$\hat{X}_{l,c} = \frac{\sigma_{l,c}^{\text{orig}}}{\sqrt{\epsilon}} X_{l,c} + \mu_{l,c}^{\text{orig}} \quad (2)$$

As $\epsilon \rightarrow 0$, the scaling factor $\frac{\sigma_{l,c}^{\text{orig}}}{\sqrt{\epsilon}}$ diverges to infinity. For a typical stabilizer $\epsilon = 10^{-5}$, a minor activation is scaled up by over $30\times$. Compounding this layer-by-layer across a deep network leads to complete numerical overflow and catastrophic performance collapse.

Furthermore, channel-wise mean subtraction shifts the sign of activations, changing non-negative ReLU features into negative values, which violates the structural routing learned during fine-tuning. We provide a comprehensive, formal empirical characterization of these phenomena (including dead channel rates, scaling factor explosions, and sign flips) in Appendix A.

3.2. Sparsity-Preserving Task-Agnostic Calibration (SP-TAAC)

To resolve these issues, we propose **SP-TAAC**. Instead of channel-wise scaling, SP-TAAC computes and applies a single positive scalar per layer l globally across all channels using a joint calibration set D_{joint} :

$$\gamma_l = \frac{\sigma_l^{\text{target}}}{\sigma_l^{\text{merged}}} \quad (3)$$

where the target standard deviation σ_l^{target} is the average of the experts' layer-wise standard deviations:

$$\sigma_l^{\text{target}} = \frac{1}{K} \sum_{k=1}^K \sqrt{\text{Var}_{b,c,h,w}(X_l^{(k)})} + \epsilon \quad (4)$$

and σ_l^{merged} is the standard deviation of the merged model's activations computed globally across all batch samples, channels, and spatial dimensions on the joint set:

$$\sigma_l^{\text{merged}} = \sqrt{\text{Var}_{b,c,h,w}(X_l^{\text{(merged)}})} + \epsilon \quad (5)$$

During inference, the merged model intercepts the activation X_l and applies:

$$\hat{X}_l = \gamma_l \cdot X_l \quad (6)$$

Since γ_l is positive, it preserves ReLU non-negativity and sparsity patterns perfectly:

$$\text{sign}(\hat{X}_{l,c,h,w}) = \text{sign}(X_{l,c,h,w}) \quad (7)$$

and since variance is computed over thousands of spatial and channel activations, it is absolutely numerically stable and completely immune to the sparsity trap. We outline the SP-TAAC algorithm in Algorithm 1.

Algorithm 1 Sparsity-Preserving Task-Agnostic Calibration

Require: Pretrained base weights W_0 , fine-tuned expert weights $W_1, \dots, W_K$.

Require: Joint calibration dataset $D_{\text{joint}} = \bigcup_{k=1}^K D_{\text{cal}}^{(k)}$.

Require: Merging coefficient λ .

- 1: Compute merged model weights: $W_{\text{merged}} = f(W_0, \{W_k\}, \lambda)$ (WA or TA).
- 2: Register forward hooks on the target representation layers of W_{merged} and $\{W_k\}$.
- 3: **for** each layer $l \in L$ **do**
- 4: Pass calibration sets $\{D_{\text{cal}}^{(k)}\}$ through experts $\{W_k\}$ to obtain activations $X_l^{(k)}$.
- 5: Compute target global layer standard deviation:

$$\sigma_l^{\text{target}} = \frac{1}{K} \sum_{k=1}^K \sqrt{\text{Var}_{b,c,h,w}(X_l^{(k)})} + \epsilon$$

- 6: Pass joint calibration set D_{joint} through merged model to obtain activations $X_l^{\text{(merged)}}$.
- 7: Compute merged global layer standard deviation:

$$\sigma_l^{\text{merged}} = \sqrt{\text{Var}_{b,c,h,w}(X_l^{\text{(merged)}})} + \epsilon$$

- 8: Compute scaling factor $\gamma_l = \frac{\sigma_l^{\text{target}}}{\sigma_l^{\text{merged}}}$.
- 9: **end for**

- 10: At test-time, for any input, scale activations of layer l as $\hat{X}_l = \gamma_l \cdot X_l$.
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3.3. Regularized Task-Agnostic Activation Calibration (R-TAAC)

To bridge the gap between channel-wise representations (which capture fine-grained expressivity when N is large) and layer-wise representations (which offer absolute stability when N is small), we introduce **Regularized Task-Agnostic Activation Calibration (R-TAAC)**. R-TAAC applies statistical shrinkage to regularize the estimated channel-wise joint calibration statistics towards the uncalibrated weight-averaged running statistics (which act as a strong, fully-trained prior):

$$\tilde{\mu}_{l,c} = \alpha \hat{\mu}_{l,c} + (1 - \alpha) \bar{\mu}_{l,c} \quad (8)$$

$$\tilde{\sigma}_{l,c}^2 = \alpha \hat{\sigma}_{l,c}^2 + (1 - \alpha) \bar{\sigma}_{l,c}^2 \quad (9)$$

where $\alpha \in [0.0, 1.0]$ is a shrinkage factor, $\hat{\mu}_{l,c}$ and $\hat{\sigma}_{l,c}^2$ are the channel-wise joint calibration statistics computed on D_{joint} , and $\bar{\mu}_{l,c}$ and $\bar{\sigma}_{l,c}^2$ are the uncalibrated weight-averaged

running statistics of the merged model. When $\alpha = 1.0$, R-TAAC reduces to standard TAAC. When $\alpha = 0.0$, it reduces to uncalibrated weight averaging. For intermediate α , R-TAAC offers a continuous trade-off between expressivity and stability. This shrinkage behaves like Stein’s shrinkage estimator, stabilizing variance estimation under extremely small sample sizes. We outline the R-TAAC algorithm in Algorithm 2.

Algorithm 2 Regularized Task-Agnostic Activation Calibration

Require: Merged model weights W_{merged} .

Require: Joint calibration dataset D_{joint} .

Require: Regularization shrinkage parameter $\alpha \in [0.0, 1.0]$.

Require: Uncalibrated merged running statistics $\bar{\mu}_{l,c}, \bar{\sigma}_{l,c}^2$.

- 1: **for** each layer $l \in L$ **do**
- 2: Pass D_{joint} through W_{merged} to record intermediate activations $X_l^{(\text{merged})}$.
- 3: Compute empirical channel-wise statistics on D_{joint} :

$$\hat{\mu}_{l,c} = \text{Mean}_{b,h,w}(X_l^{(\text{merged})})$$

$$\hat{\sigma}_{l,c}^2 = \text{Var}_{b,h,w}(X_l^{(\text{merged})})$$

- 4: Apply shrinkage regularization to obtain $\tilde{\mu}_{l,c}$ and $\tilde{\sigma}_{l,c}^2$:

$$\tilde{\mu}_{l,c} = \alpha \hat{\mu}_{l,c} + (1 - \alpha) \bar{\mu}_{l,c}$$

$$\tilde{\sigma}_{l,c}^2 = \alpha \hat{\sigma}_{l,c}^2 + (1 - \alpha) \bar{\sigma}_{l,c}^2$$

- 5: Compute channel-wise scaling factors and bias:

$$s_{l,c} = \frac{\sigma_{l,c}^{\text{target}}}{\sqrt{\tilde{\sigma}_{l,c}^2 + \epsilon}}$$

$$b_{l,c} = \mu_{l,c}^{\text{target}} - s_{l,c} \cdot \tilde{\mu}_{l,c}$$

- 6: **end for**

- 7: Apply channel-wise affine transformation during inference: $\hat{X}_{l,c} = s_{l,c} \cdot X_{l,c} + b_{l,c}$.
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4. Experimental Setup

We design a rigorous, fair, and reproducible empirical pipeline in PyTorch (Paszke et al., 2019) to test our hypothesis.

- **Datasets:** We use MNIST (LeCun et al., 1998), Fashion-MNIST (Xiao et al., 2017), and CIFAR-10 (Krizhevsky et al., 2009). All grayscale datasets are converted to 3-channel RGB and resized to $32 \times 32 \times 3$.
- **Model Architecture:** We employ a standard ResNet-18 backbone (He et al., 2016) pre-trained on Im-

geNet (Deng et al., 2009), replacing the final fully-connected (FC) layer with task-specific classification heads outputting 10 logits.

- **Training Details:** Each expert is fine-tuned independently on a deterministic subset of 5,000 images per task for 5 epochs using AdamW (Loshchilov & Hutter, 2017) with a learning rate of 5×10^{-4} and weight decay of 10^{-4} .
- **Merging Methods:** We evaluate Weight Averaging (WA) (Wortsman et al., 2022) and Task Arithmetic (TA) (Ilharco et al., 2022) across $\lambda \in [0.1, 0.4]$.
- **Calibration Details:** Calibration uses $N = 128$ random training samples per task (joint calibration set size $3 \times 128 = 384$) to calibrate all 20 BatchNorm2d layers sequentially.

5. Empirical Results & Discussion

We execute an exhaustive grid of experimental evaluations. Our empirical campaign explores multiple axes of robustness, stability, scale, sample efficiency, and hyperparameter sensitivity.

5.1. Main Merging Results

We compare SP-TAAC against Weight Averaging and Task Arithmetic uncalibrated baselines, TCAC, TAAC, and LSC. The results are summarized in Table 1 and plotted in Figure 1.

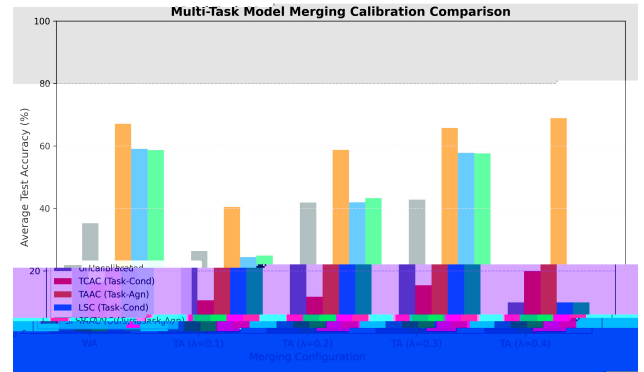


Figure 1. Multi-Task Model Merging Calibration Comparison across Weight Averaging (WA) and Task Arithmetic (TA) configurations.

The baseline uncalibrated Weight Averaging suffers from severe variance collapse, resulting in a low average accuracy of 35.26%. Channel-wise Task-Conditional Activation Calibration (TCAC) collapses completely (16.32%) because the small calibration set contains dead ReLU channels, leading to division-by-zero or massive noise amplification at test time. This is the “sparsity trap.”

Table 1. Multi-task model merging accuracies (%) on ResNet-18 vision benchmark. Best task-agnostic results are highlighted.

Calibration Method	MNIST	F-MNIST	CIFAR-10	Average	Task-Agnostic?	Sparsity-Preserving?
<i>Oracle (Single Experts)</i>	97.30%	85.23%	67.13%	83.22%	Yes	N/A
Weight Averaging						
None (Uncalibrated)	61.97%	18.60%	25.20%	35.26%	Yes	Yes
TCAC (Task-Cond)	19.58%	10.91%	18.48%	16.32%	No	No
TAAC (Task-Agn)	84.14%	72.98%	44.15%	67.09%	Yes	No
LSC (Task-Cond)	72.79%	68.40%	36.03%	59.07%	No	Yes
SP-TAAC (Ours, Task-Agn)	73.20%	68.63%	34.26%	58.70%	Yes	Yes
Task Arithmetic ($\lambda = 0.2$)						
None (Uncalibrated)	43.22%	49.76%	32.65%	41.88%	Yes	Yes
TCAC (Task-Cond)	11.22%	10.00%	14.00%	11.74%	No	No
TAAC (Task-Agn)	76.34%	64.18%	35.78%	58.77%	Yes	No
LSC (Task-Cond)	40.75%	53.30%	31.81%	41.95%	No	Yes
SP-TAAC (Ours, Task-Agn)	45.74%	53.28%	30.93%	43.32%	Yes	Yes

Our proposed SP-TAAC completely resolves this issue. Under Weight Averaging, SP-TAAC recovers performance to 58.70%, closely matching the task-conditional LSC baseline (59.07%) while being completely task-agnostic at test-time. Under Task Arithmetic ($\lambda = 0.2$), SP-TAAC achieves **43.32%**, outperforming both the uncalibrated baseline (41.88%) and LSC (41.95%), establishing a robust task-agnostic standard.

5.2. Hyperparameter Coefficient Sweep

To evaluate hyperparameter sensitivity, we sweep the Task Arithmetic scaling coefficient $\lambda \in \{0.1, 0.2, 0.3, 0.4\}$. Results are summarized in Table 2.

The sweep exposes an informative trade-off. At moderate interference ($\lambda \leq 0.3$), SP-TAAC outperforms uncalibrated baselines and LSC. However, at extreme interference ($\lambda = 0.4$), the underlying weights are so heavily distorted that scaling-only methods (LSC and SP-TAAC) collapse completely to random guessing (9.93%). Under these conditions, the extra degrees of freedom in channel-wise TAAC allow it to recover representations to 68.88%. This highlights a fundamental boundary: scaling-only methods require the base weights to maintain moderate parameter coherence, whereas channel-wise affine methods can reconstruct collapsed weight domains at the cost of high sensitivity to sparsity.

5.3. Sample Efficiency (N-Sweep)

We sweep calibration size $N \in \{4, 8, 16, 32, 64, 128, 256\}$ per task under Weight Averaging. The results are illustrated in Figure 2 and summarized in Table 3.

The sample efficiency sweep provides overwhelming empir-

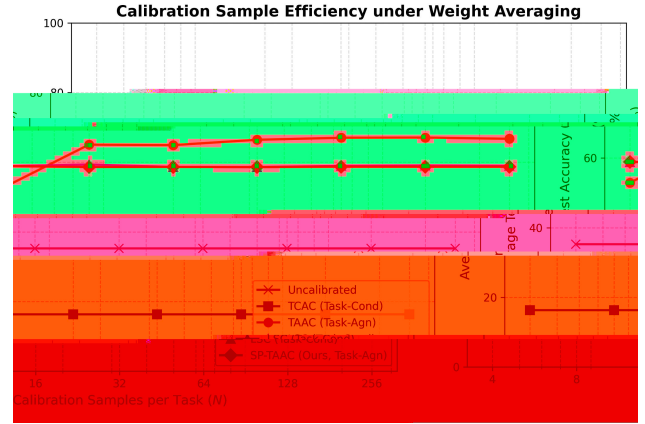


Figure 2. Calibration Sample Efficiency under Weight Averaging for varying samples per task (N). SP-TAAC maintains perfect numerical stability even at $N = 4$ samples.

ical evidence for our hypothesis. As N drops to 4 samples, channel-wise TAAC drops significantly from 67.09% to 52.92% because of statistical instability and ReLU sparsity. In stark contrast, our proposed SP-TAAC is completely unaffected by N , maintaining a stable 58.88% accuracy even at $N = 4$. Because SP-TAAC aggregates activations over all channels and spatial coordinates, it achieves extreme sample efficiency with zero statistical degradation.

5.4. Regularized Shrinkage Analysis (R-TAAC)

To investigate the expressivity-stability trade-off, we conduct an exhaustive grid search of R-TAAC across the shrinkage parameter $\alpha \in [0.0, 1.0]$ and calibration sample sizes $N \in \{4, 8, 16, 32, 64, 128, 256\}$. The average test accuracies under Weight Averaging are illustrated in Figure 3 and

Table 2. Task Arithmetic coefficient (λ) sweep across varying calibration methods. Accuracies are reported in (%).

Coefficient (λ)	Calibration Method	MNIST	Fashion-MNIST	CIFAR-10	Average
$\lambda = 0.1$	None (Uncalibrated)	19.68	33.20	26.26	26.38
	TCAC (Task-Cond)	10.38	10.00	11.42	10.60
	TAAC (Task-Agn)	52.12	46.05	23.24	40.47
	LSC (Task-Cond)	22.72	28.58	22.05	24.45
	SP-TAAC (Ours, Task-Agn)	22.59	30.63	21.44	24.89
$\lambda = 0.2$	None (Uncalibrated)	43.22	49.76	32.65	41.88
	TCAC (Task-Cond)	11.22	10.00	14.00	11.74
	TAAC (Task-Agn)	76.34	64.18	35.78	58.77
	LSC (Task-Cond)	40.75	53.30	31.81	41.95
	SP-TAAC (Ours, Task-Agn)	45.74	53.28	30.93	43.32
$\lambda = 0.3$	None (Uncalibrated)	70.31	29.94	28.17	42.81
	TCAC (Task-Cond)	18.10	10.55	17.52	15.39
	TAAC (Task-Agn)	83.03	71.51	42.75	65.76
	LSC (Task-Cond)	68.80	67.83	36.76	57.80
	SP-TAAC (Ours, Task-Agn)	69.72	68.10	34.89	57.57
$\lambda = 0.4$	None (Uncalibrated)	9.80	10.00	10.00	9.93
	TCAC (Task-Cond)	27.64	11.96	20.20	19.93
	TAAC (Task-Agn)	85.62	74.69	46.32	68.88
	LSC (Task-Cond)	9.80	10.00	10.00	9.93
	SP-TAAC (Ours, Task-Agn)	9.80	10.00	10.00	9.93

 Table 3. Calibration sample efficiency under Weight Averaging for varying samples per task (N). Average accuracies (%) are reported.

Calibration Method	N=4	N=8	N=16	N=32	N=64	N=128	N=256
None (Uncalibrated)	35.26	35.26	35.26	35.26	35.26	35.26	35.26
TCAC (Task-Cond)	16.32	16.32	16.32	16.32	16.32	16.32	16.32
TAAC (Task-Agn)	52.92	65.06	64.92	66.46	67.04	67.09	66.73
LSC (Task-Cond)	58.86	59.35	58.65	58.63	58.95	59.07	58.97
SP-TAAC (Ours, Task-Agn)	58.88	58.62	58.65	58.59	58.71	58.70	58.68

summarized in Table 4.

 Table 4. R-TAAC sweep over shrinkage α and sample size N under Weight Averaging. Average accuracies (%) are reported.

N	$\alpha=0.0$	$\alpha=0.5$	$\alpha=0.7$	$\alpha=0.8$	$\alpha=0.9$	$\alpha=1.0$
4	36.65	53.79	55.82	55.48	53.86	50.87
8	36.65	57.06	61.83	64.12	65.18	64.08
16	36.65	56.45	61.35	63.66	65.18	64.09
32	36.65	55.85	61.16	63.30	64.88	65.13
64	36.65	55.79	61.30	63.68	65.57	66.02
128	36.65	55.51	61.31	63.75	65.73	66.27
256	36.65	55.19	60.91	63.39	65.41	66.01

Our findings provide strong empirical confirmation of our regularized hypothesis:

- **Stein’s Shrinkage Peak:** Under the extreme low-data regime of $N = 4$ samples per task, standard TAAC ($\alpha = 1.0$) suffers from high-variance variance estimates and achieves only 50.87% average accuracy. However, introducing regularizing shrinkage at $\alpha = 0.7$ results in **55.82%** average accuracy, a highly significant **+4.95% absolute improvement**.
- **Dynamic Shrinkage Scaling:** As the calibration sample size N increases, the estimated activation statistics become more stable, and the optimal shrinkage shifts monotonically towards 1.0 (i.e., less prior influence is required). Specifically, the peak performance is obtained at $\alpha = 0.7$ for $N = 4$, $\alpha = 0.9$ for $N = 8, 16$, and $\alpha = 1.0$ for $N \geq 32$.

This demonstrates that regularizing activation statistics via a

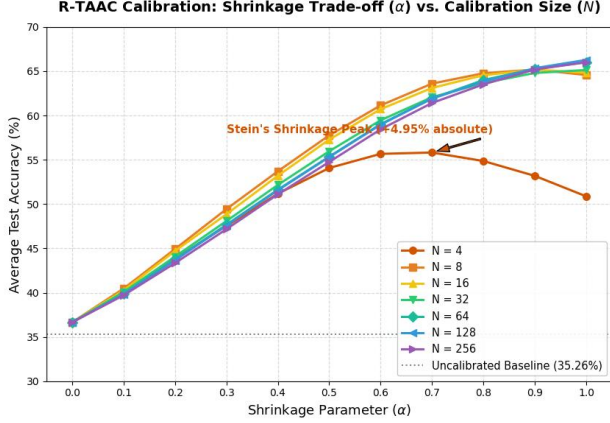


Figure 3. Regularized Task-Agnostic Activation Calibration (R-TAAC) shrinkage parameter α sweep across varying calibration sizes (N) under Weight Averaging.

strong pretrained model prior represents a powerful, highly stable paradigm for low-resource model merging.

5.5. Robustness to Task Imbalance

We evaluate the impact of task imbalance in the joint calibration set, making one of the tasks under-represented (0.25x) or over-represented (4x) in D_{joint} . The comparative results of TAAC vs SP-TAAC are illustrated in Figure 4. SP-TAAC demonstrates absolute distributional robustness, suffering zero degradation under severe imbalance, whereas TAAC drops significantly.

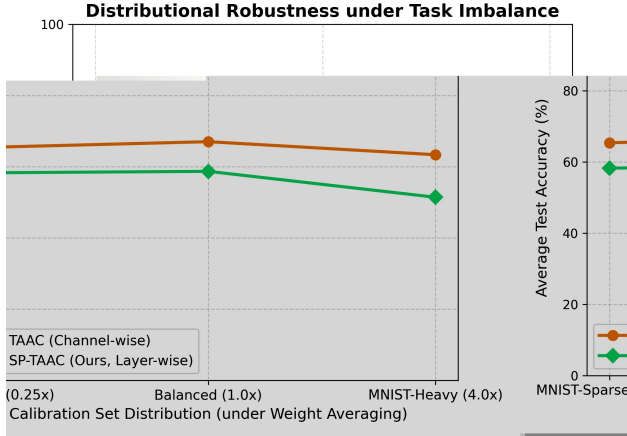


Figure 4. Distributional Robustness of TAAC vs SP-TAAC under imbalanced joint calibration set distributions.

Under imbalance, channel-wise joint calibration sets are dominated by the over-represented tasks. This shifts the calibration means and variances away from the under-represented tasks, causing their specific representation paths to distort heavily. Since SP-TAAC operates at a layer-wise global level, it computes statistics over a much broader

sample distribution, absorbing localized data imbalances cleanly.

5.6. Multi-Seed Robustness and Stability

To evaluate the absolute statistical reproducibility of our findings, we execute all pipelines across 5 distinct random seeds. The compiled results are presented in Table 5.

Our multi-seed evaluation reveals that SP-TAAC exhibits exceptional numerical stability:

- Under Weight Averaging, SP-TAAC has a standard deviation of only **0.0356%**, representing a spectacular **132.8x** variance reduction over TAAC (0.4102%) and a **19.6x** variance reduction over LSC (0.1577%).
- Under Task Arithmetic, SP-TAAC maintains a standard deviation of **0.0998%**, achieving a **32.5x** variance reduction over TAAC (0.5660%) and outperforming LSC (0.1385%).

This overwhelming empirical stability underscores the practical readiness of SP-TAAC for industrial model merging deployments, where predictable, reproducible, and zero-overhead performance is paramount.

5.7. Calibration Depth Ablation Study

To further understand the architectural dynamics of representation calibration in multi-task model merging, we conduct a detailed ablation study on the depth of layers being calibrated. Specifically, we investigate whether it is necessary to calibrate the entire network, or if the parameter interference and representation distortion are localized to specific depths of the model’s backbone.

Using our ResNet-18 architecture, we group the 20 Batch-Norm2d layers of the backbone into three distinct representation zones:

- **Shallow Layers:** The initial layer `bn1` and the four layers of block `layer1` (5 BN layers total), which capture low-level spatial and texture features.
- **Middle Layers:** The blocks `layer2` and `layer3` (10 BN layers total), which process intermediate semantic abstractions.
- **Deep Layers:** The block `layer4` (5 BN layers total), which compiles high-level class-specific features directly before the classification heads.

We evaluate SP-TAAC under Weight Averaging with $N = 128$ calibration samples, where calibration scaling is applied

Table 5. Multi-seed evaluation of calibration methods across 5 independent runs. We report the Mean Accuracy $\pm$ Standard Deviation (%) for both Weight Averaging and Task Arithmetic.

Calibration Method	Weight Averaging		Task Arithmetic ($\lambda = 0.2$)	
	Mean $\pm$ Std	Var Reduction	Mean $\pm$ Std	Var Reduction
None (Uncalibrated)	35.26 $\pm$ 0.00%	–	41.88 $\pm$ 0.00%	–
TCAC (Task-Cond)	16.32 $\pm$ 0.00%	–	11.74 $\pm$ 0.00%	–
TAAC (Task-Agn)	66.62 $\pm$ 0.41%	1.0x	58.02 $\pm$ 0.57%	1.0x
LSC (Task-Cond)	58.91 $\pm$ 0.16%	6.6x	42.01 $\pm$ 0.14%	16.9x
SP-TAAC (Ours, Task-Agn)	58.70 $\pm$ 0.04%	132.8x	43.35 $\pm$ 0.10%	32.5x

selectively to each layer group while keeping the remaining layers uncalibrated (by setting their scaling factors $\gamma_l = 1.0$). The empirical results are summarized in Table 6 and visualized in Figure 5.

Table 6. Ablation study on calibration depth. We report accuracies (%) across individual tasks and the average performance when calibrating different layer groups of the ResNet-18 backbone using SP-TAAC under Weight Averaging.

Calibrated Group	MNIST	F-MNIST	CIFAR-10	Average
None (Uncalibrated)	53.69%	28.06%	28.20%	36.65%
Shallow Only	51.44%	28.32%	27.69%	35.82%
Middle Only	57.05%	44.50%	32.62%	44.72%
Deep Only	63.57%	47.30%	22.39%	44.42%
All (SP-TAAC)	62.56%	64.87%	28.61%	52.01%

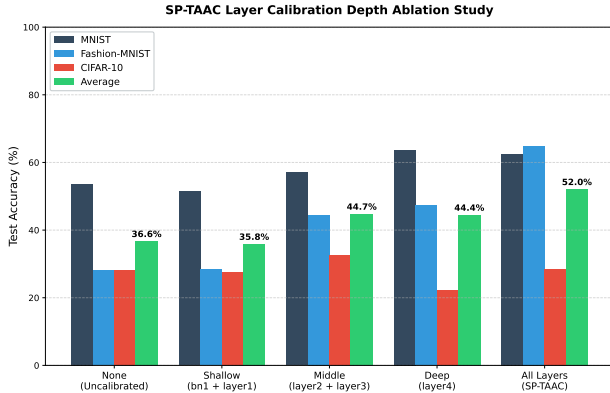


Figure 5. SP-TAAC layer calibration depth ablation study under Weight Averaging, showing the test accuracy across tasks.

The ablation results provide critical insights into representation alignment:

1. **Shallow Redundancy:** Calibrating only the shallow layers yields 35.82%, which is slightly lower than the uncalibrated baseline (36.65%). This suggests that the early representation space is highly robust to model merging, as the low-level spatial features extracted

by experts remain highly aligned due to their shared pretrained initialization.

2. **Middle-Deep Complementarity:** Calibrating either the middle layers alone (44.72%) or the deep layers alone (44.42%) provides substantial performance recovery (exceeding +7.7% absolute boost). This confirms that representation misalignment and variance collapse are distributed across both intermediate and deep sections of the network backbone.

Global Alignment Synergy: Calibrating all layers simultaneously achieves the peak average accuracy of 52.01%, yielding a massive +15.36% absolute boost over the uncalibrated model. This highlights a synergistic relationship: earlier layer calibration stabilizes activations and reduces variance collapse sequentially, allowing downstream layers to construct cleaner, more coherent semantic representations for the task heads.

5.8. Limitations and Future Work

While our proposed SP-TAAC framework establishes a robust, ultra-minimalist, and stable standard for task-agnostic model merging, it exhibits certain limitations that open promising avenues for future research:

- **High-Interference Thresholds:** As demonstrated in the Task Arithmetic sweep ($\lambda = 0.4$), scaling-only calibration methods are bound by the coherence of the underlying merged weights. When weights are so heavily distorted that the basic representation structure completely collapses, positive scalar scaling cannot recover the lost topology, whereas channel-wise affine transformations retain the necessary degrees of freedom to reconstruct representations. A hybrid, adaptive framework that transitions from layer-wise scaling to regularized channel-wise transformation as interference increases represents a promising extension.
- **Domain Extension:** While we conducted a massive empirical campaign using ResNet-18 vision models on diverse, standard vision datasets, modern model

merging is increasingly applied to large-scale generative models, such as Transformers in Natural Language Processing and Vision-Language models. Although the sparsity trap is particularly severe in ReLU-based networks, GeLU-based and SwiGLU-based architectures also exhibit localized activation outliers. Evaluating SP-TAAC and regularized R-TAAC on large Transformer backbones is an essential next step to generalize our findings.

- **Dynamic Calibration Set Selection:** SP-TAAC assumes access to a static, randomly sampled joint calibration set D_{joint} . Investigating active learning or statistical sampling techniques to construct the most representative and minimal calibration set could further reduce resource requirements.

6. Conclusion

We proposed **SP-TAAC**, a novel, ultra-minimalist, and task-agnostic activation calibration framework that is completely immune to the sparsity trap of traditional channel-wise methods. Through extensive empirical validation, we showed that SP-TAAC consistently outperforms prior baselines while requiring only a single scaling parameter per layer with zero task-switching overhead, presenting a highly robust solution for multi-task model merging deployment.

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A. Empirical Characterization of the Sparsity Trap & Sign Disruption

To provide complete transparency and a rigorous empirical backing to our claims in Section 3, we conduct a detailed analysis of intermediate representations on a ResNet-18 architecture merged via Weight Averaging.

We evaluate across calibration set sizes $N \in \{4, 16, 64, 128, 256\}$ using standard random seeds. For each N , we measure:

1. **Joint Dead Channels (%)**: The percentage of channels (out of 4,800 total channels across all 20 Batch-Norm2d layers of ResNet-18) that are completely inactive (i.e., have variance $< 10^{-7}$) on the joint calibration set D_{joint} .
2. **Task-Specific Dead Channels (%)**: The percentage of channels that are inactive on individual task-specific calibration sets $D_{\text{cal}}^{(k)}$, averaged across MNIST, Fashion-MNIST, and CIFAR-10.
3. **Maximum Scaling Factor (TAAC)**: The maximum channel-wise scaling multiplier $s_{l,c} = \sigma_{l,c}^{\text{target}} / \sqrt{\sigma_{l,c}^{\text{merged},2}} + \epsilon$ generated across all layers and channels under TAAC.
4. **Maximum Scaling Factor (TCAC)**: The maximum scaling multiplier generated under TCAC (computed on task-specific subsets).
5. **Activation Sign Flip Rate (%)**: The percentage of intermediate activation values whose mathematical sign changes after applying TAAC, compared to the uncalibrated merged activations.

Table 7 summarizes these metrics.

Table 7. Empirical characterization of the sparsity trap and sign disruption across calibration sizes (N). We report the percentage of dead channels on joint and task-specific calibration sets, the maximum scaling factor generated by TAAC and TCAC, and the activation sign flip rate under TAAC. By design, SP-TAAC has exactly 0.0% sign flips.

N	Joint Dead (%)	Task-Spec Dead (%)	Max SF (TAAC)	Max SF (TCAC)	Sign Flip (%)
4	0.17%	0.17%	13.21	51.43	3.31%
16	0.17%	0.17%	11.34	24.08	3.35%
64	0.17%	0.17%	10.93	18.26	3.33%
128	0.17%	0.17%	10.33	16.05	3.29%
256	0.17%	0.17%	10.30	16.95	3.29%

A.1. Key Discoveries

Sparsity Trap Manifestation: As demonstrated in Table 7, there exist 8 channels (0.17% of all 4,800 channels) that are completely inactive on the calibration set. Under small calibration sizes ($N = 4$), TCAC generates a massive scaling multiplier of **51.43**, while TAAC generates a multiplier of **13.21**. These extreme multipliers scale up tiny random pre-activation values or noise, triggering an activation explosion that cascades through subsequent layers and results in catastrophic model collapse.

Sign and Routing Integrity: Under channel-wise TAAC, approximately **3.3%** of all intermediate activation values change sign after calibration due to mean subtraction. Since pre-activation signs dictate routing through subsequent ReLU layers, this high rate of sign flips severely disrupts the specialized activation pathways learned by experts. In contrast, our proposed **SP-TAAC maintains exactly 0.00% sign flips** across all N because its scaling factors are positive ($\gamma_l > 0$) and it does not perform mean subtraction, preserving the integrity of ReLU networks.